

Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Unex "Metropolis" Style
Supplied to:	Auckland Council
Client / designers name:	
Description of work: <i>E.g. New dwelling</i>	
Address of property:	

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☐ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ <i>please specify</i>	
Location of deck / balcony / stairs:	☐ Internal	☐ External	☐ <i>please specify</i>			
Deck / balcony / stairs:	☐ New	☐ Existing	Cladding:	☐ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☐ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer
Thickness:	(mm)	Width of pane:	(m)
Height above floor:	(m)	Height above fixing:	(m)
Note: If barrier extends more than 1.2m above fixing point, test reports must be provided			
Supplier:	Royal Glass		

PART 4: BARRIER DETAILS

System name:	Unex "Metropolis" Style
Drawings (ref page number)	Technical Catalogue pages: SS.60.12T, FS.2T.22.60, AS.60.12T
Engineers name:	Dale Donovan
Hardware (type):	Aluminium posts, and glass clamps

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES (tick all boxes that apply)☐ Continuous interlinking rail (show how fixed to building and each pane of glass)☒ Continuous interlinking top cap (show how fixed to building and each pane of glass)☐ Clamps, brackets or point fixings (show how fixed to building and each pane of glass)☐ Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass)☐ Glass with stiff interlayer☐ Two or three glass edges supported by continuous clamps☒ Structural post to fix interlinking rail to deck☐ Other (specify)

Engineers name:

Dale Donovan

Drawings (ref page number)

Comments:

PART 6: COVER SHEET COMPLETED BY

Name:

Roxy Huang

Signature:



Date:

Contact number:

0800 769 254

Email:

info@royalglass.co.nz

BALUSTRADE PS1 COVER SHEET

Date:

Product | Fixture:

Metropolis(VRRTop) | FS.2T.22.60-WetTimber,LVL
Beams,Screws

Indicative Linear Meterage:

Project Name:

Site Address:

Legal Description:

SPECIFIER DETAILS

**Area where balustrade is
installed:**

Company:

Royal Glass

Phone:

0800 769 254

Email:

support@royalglass.co.nz

Website:

www.royalglass.co.nz

NOTE: The attached PS1 covers the relevant balustrade system for multiple sites and applications. The engineer's signature date relates to the calculation of this data. The effective date of this individual PS1 is as shown on this cover sheet.

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UNEX
ARCHITECTURAL BARRIERS

BALUSTRADES SCREENS POOL FENCES HANDRAILS



association of
consulting and
engineering



PRODUCER STATEMENT – PS1 DESIGN

BUILDING CODE CLAUSE(S): B1/VM1, F2, F4

JOB NUMBER:

ISSUED BY: DODD CIVIL CONSULTANTS

(Engineering Design Firm)

TO:

(Owner/Developer)

TO BE SUPPLIED TO: TO WHOM IS MAY CONCERN

(Building Consent Authority)

IN RESPECT OF: THE SUPPLY OF UNEX ALUMINIUM AND GLASS BALUSTRADE FOR THE ADDRESS BELOW

(Description of Building Work)

AT:

(Address, Town/City)

LEGAL DESCRIPTION:

N/A ☐

We have been engaged by the owner/developer referred to above to provide (Extent of Engagement):

THE SUPPLY OF UNEX ALUMINIUM AND GLASS BALUSTRADE IN THE METROPOLIS STYLE ONLY

in respect of the requirements of the Clause(s) of the Building Code specified above for Part only, as specified in the Schedule, of the proposed building work.

The design carried out by us has been prepared in accordance with:

- ☒ Compliance documents issued by the Ministry of Business, Innovation & Employment (Verification method/acceptable solution) B1/VM1, F2, F4 and/or;
- ☐ Alternative solution as per the attached Schedule.

The proposed building work covered by this producer statement is described on the drawings specified in the Schedule, together with the specification, and other documents set out in the Schedule.

On behalf of the Engineering Design Firm, and subject to:

- Site verification of the following design assumptions:
- All proprietary products meeting their performance specification requirements;

I believe on reasonable grounds that:

- the building, if constructed in accordance with the drawings, specifications, and other documents provided or listed in the Schedule, will comply with the relevant provisions of the Building Code and that;
- the persons who have undertaken the design have the necessary competency to do so.

I recommend the CM 2 level of **construction monitoring**.

I, (Name of Engineering Design Professional) DALE DONOVAN, am:

- ☒ CPEng number 1004603

and hold the following qualifications BMS, BE(CIVIL)

The Engineering Design Firm holds a current policy of Professional Indemnity Insurance no less than \$200,000

The Engineering Design Firm is not a member of ACE New Zealand.

SIGNED BY (Name of Engineering Design Professional): DALE DONOVAN

(Signature below):

ON BEHALF OF (Engineering Design Firm): DODD CIVIL CONSULTANTS

Date: 25 AUG 2023

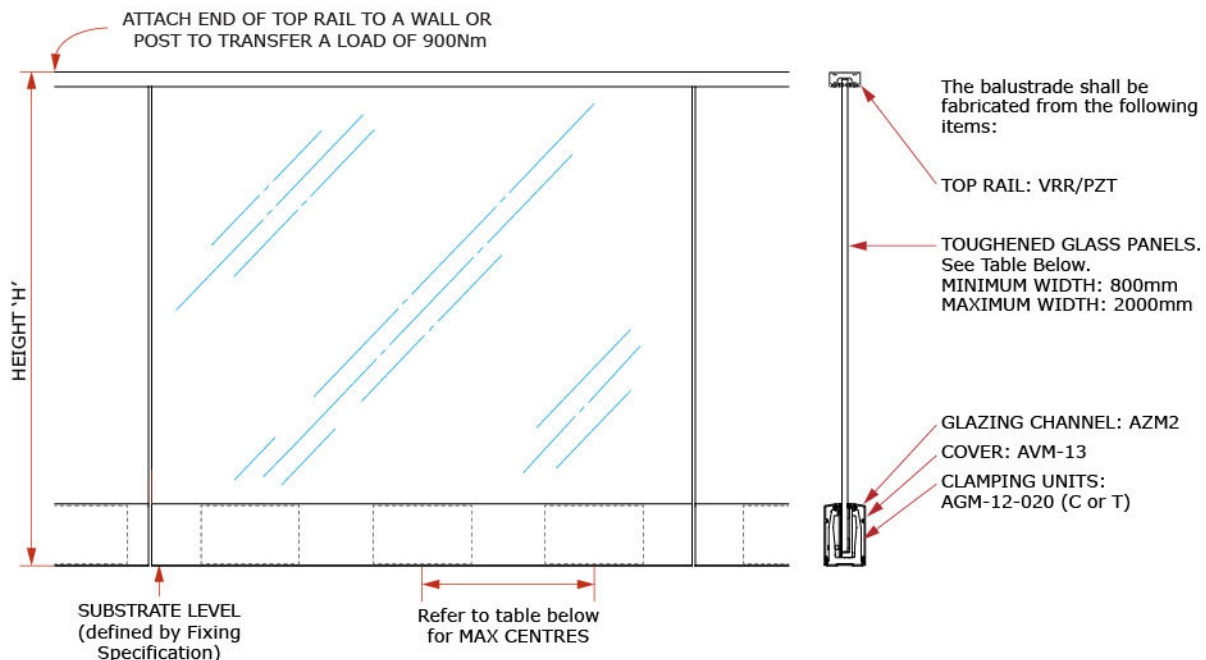
Note: This statement has been prepared solely for the Building Consent Authority named above and shall not be relied upon by any other person or entity. Any liability in relation to this statement accrues to the Engineering Design Firm only. As a condition of reliance on this statement, the Building Consent Authority accepts that the total maximum amount of liability of any kind arising from this statement and all other statements provided to the Building Consent Authority in relation to this building work, whether in tort or otherwise, is limited to the sum of \$200,000.

This form is to accompany **Form 2 of the Building (Forms) Regulations 2004** for the application of a Building Consent.

'METROPOLIS' (VRR TOP RAIL)

This is a Style Specification for the Metropolis® Balustrade system. It must only be used in conjunction with a Fixing Specification from pages 85-89, and the Assembly Specification AS.60.12T on Page 102. Installation must be completed in accordance with the Installation Guide on pages 168-171 and all other relevant portions of this manual.

1. Glass shall be **GRADE A SAFETY GLASS, TOUGHENED** in accordance with AS/NZS 2208, with a minimum thickness as determined from Table below. Glass shall be supported and glazed in accordance with NZS 4223. All exposed edges to be flat polished.
2. The top rail shall be attached to the top edge of the glass with structural silicone, for the full length of the rail and for the full width of the glass. The top rail ends shall be structurally attached to a building element to transfer a load of 900 Newtons in any direction. At any joins or corners, the top rail shall be structurally gusseted to transfer a load of 900 Newtons in any direction.
3. A Top Rail is not required on the Frameless Glass styles if not preventing a fall of 1.0m or more.



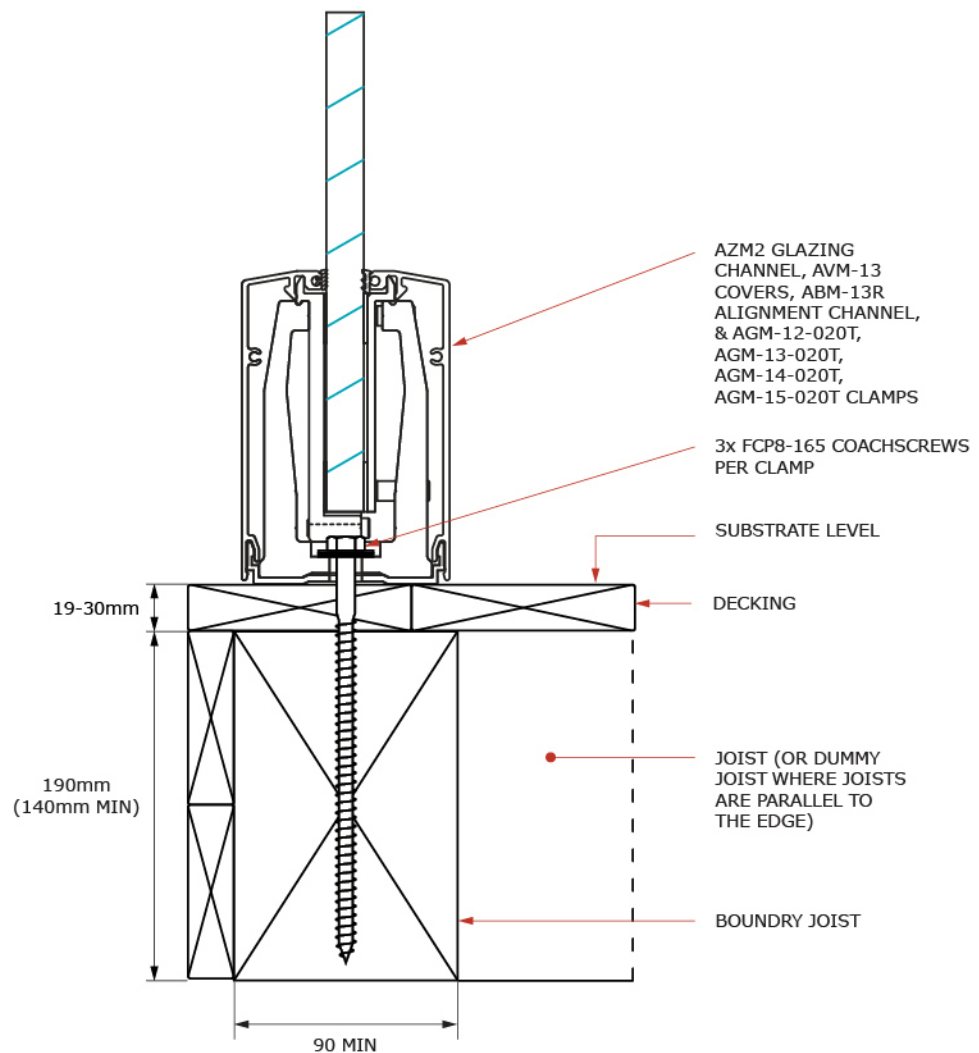
MINIMUM GLASS THICKNESS (MILLIMETRES) & CLAMP CENTRES

HEIGHT ⁽²⁾	Description ⁽⁴⁾	Line No.	LOADING CLASS ⁽¹⁾																		
			N07C/N07R									N03R	Not Preventing Falls								
			Design Wind Speed ⁽³⁾										Design Wind Speed ⁽³⁾								
			VH			EH							M			H			VH		
50	52	54	56	58	60	62	64		N/A	38	40	42	44	46	48	50	52	54	56		
1.0	Maximum Clamping Set Centres (12mm Toughened)	1	400	400	400	400	400	400	380	360	400	400	400	400	400	400	400	400	400	400	
1.1	Maximum Clamping Set Centres (12mm Toughened)	2	400	400	400	380	360	340	320	300	400	400	400	400	400	400	400	400	400	380	
1.2	Maximum Clamping Set Centres (12mm Toughened)	3	400	380	340	320	300	*	*	*	400	400	400	400	400	400	400	380	340	320	

1. **LOADING CLASS:** Refer to Page 180 for the scope of the Loading Class designations.
 2. **HEIGHT 'H':** is the overall height of the balustrade above the substrate level shown. Interpolate for Heights between those shown.
 3. **DESIGN WIND SPEED:** in m/s, Refer to Pages 37-38 for details of applicable wind codes and the methods for determining the Design Wind Speed.
 4. **CLAMPING SET CENTRES:** Figures given are the maximum spacing in mm. Refer to Chapter 5 for method of determining actual spacing required.
- * These wind speeds are achievable however they require Site Specific Engineering Designs

'METROPOLIS' - WET TIMBER, SCREWS

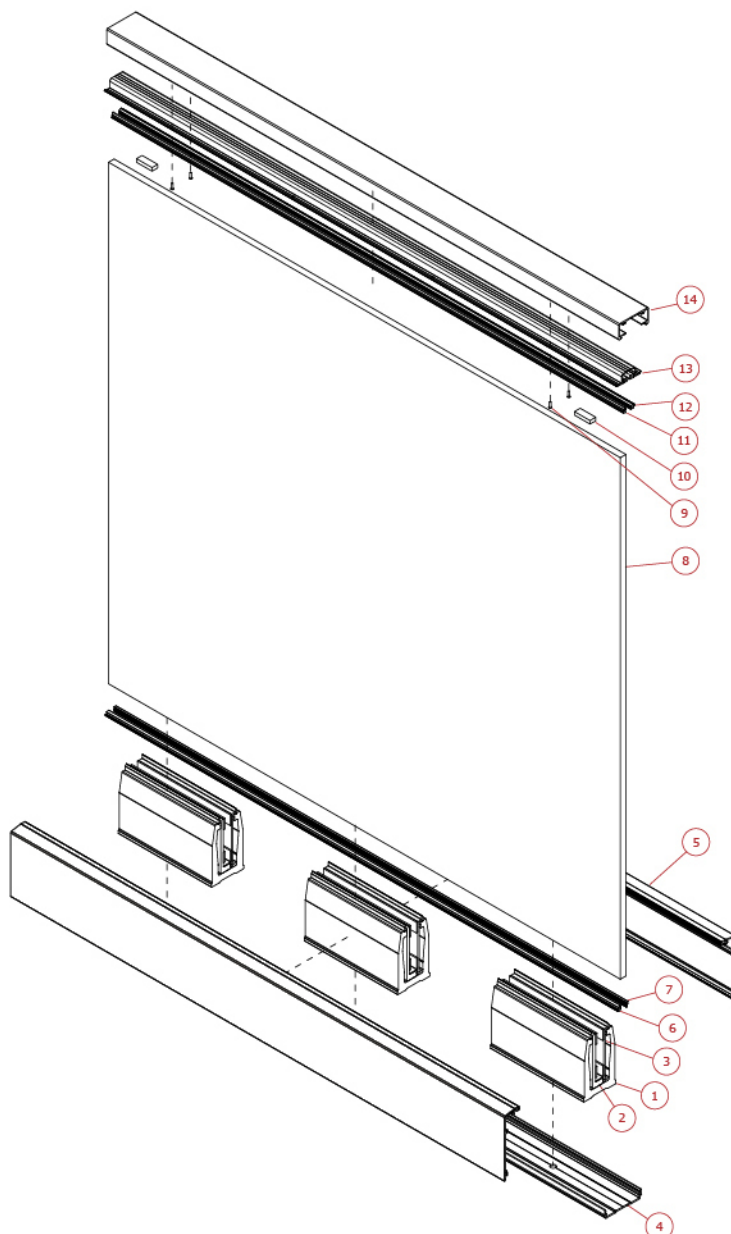
This is a Standard Fixing Specification for the 'Metropolis' Styles to the timber substrates using the AZM2 channel. It must only be used in conjunction with the relevant Style Specifications on pages 41-53. Installation must be completed in accordance with the Installation Guide on pages 168-171 and all other relevant portions of this manual.



NOTES RELATING TO THIS SPECIFICATION

1. Structural timber must be Pinus Radiata VSG8 or MSG8, or higher, see General Notes on page 80.
2. Fasteners shall be positioned on the center groove line. The 90mm wide timber joist, can be a structurally laminated timber beam, but it **cannot** consist of two x 45mm wide timber joists.
3. Washers to be fitted under coachscrews shall be stainless steel & polymer washers.
4. A level and uniform surface must be provided beneath the Glazing Channel. The Glazing Channel is not recommended to be installed over tiles.
5. Fabrication and Installation are to be in accordance with the relevant requirements of the Assembly Specifications in Chapter 4, the Installation Guides in Chapter 5, and other applicable portions of this Frameless Glass Manual.
6. Substrate design including waterproofing and the structural design of the timber substrate and its connections are not included in this specification and must be carried out by others.
7. The 3x coachscrews illustrated may be replaced with 2x M8 threaded rod extending and bolting right through the joists. Washer to be stainless steel & EPDM with FN8 nut on top, 50mm square stainless steel washers, stainless steel washers, and nyloc nut (FN8N) on the underside.

'METROPOLIS' (VRR TOP RAIL)



Refer elsewhere for corners, slopes and other situations not illustrated here. Additional spacers and gaskets between channels and substrate may be required in some situations

ITEM	Part Number	Description
1	AGM-00-020X	Structural block assembly for AZM2 system. Clamps glass to ABM-13R channel to vertically adjust when leveling
2	ABB	Bearing block. Part of AGM-00-020X
3	ABP	Bearing plate. Part of AGM-00-020X
4	ABM-13R	Pre-drilled alignment channel
5	AVM-13	Cover extrusion for on both sides
6	SRG40	Retained gasket, SRG40(12mm) - SRG30 (13.5mm & 14.28mm)/SRG20 (15mm)
7	SWE35	Wedge, SWE35(12mm) - SWE30 (13.5mm)/SWE25 (14.28mm & 15mm)
8	[Glass Infill]	Monolithic Toughened Safety Glass, Refer to Style Specification for appropriate thickness. Exposed edges Flat Polished.
9	FS4-4	Inner Rail rivets, 1 pair 150mm from ends and then 1000mm maximum centres between. Drill size = Ø3.3mm.
10	SBC	Setting blocks at 500mm centres to prevent over-compression of bonding silicone and protect glass during rail placement
11	SRG40	Retained gasket, SRG40(12mm) - SRG30 (13.5mm & 14.28mm)/SRG20 (15mm)
12	SWE35	Wedge, SWE35(12mm) - SWE30 (13.5mm)/SWE25 (14.28mm & 15mm)
13	PZT	Inner Top Rail, Structural silicone bond to top edge of glass
14	VRR	Top Rail

UNEX LOADING CLASSIFICATION

Below is an extract from AS/NZS 1170.1:2002, a standard which is cited in the NZBC. This table gives the minimum imposed actions on barriers for various occupancy types. To the right hand side of this table are the applicable UNEX Loading Classifications (ULC) which are stated in the specifications contained in this catalogue.

AS/NZS 1170.1:2002 TABLE 3.3 MINIMUM IMPOSED ACTIONS FOR BARRIERS								ULC (UNEX Loading Classifi- cations)
Type of occupancy for part of the building or structure	Specific Uses	Top Edge			Infill			
		Horizontal kN/m	Vertical kN/m	Inwards, Outwards, or Downwards kN	Horizontal kPa	Any Direction kN		
A Domestic and residential activities	All areas within or serving exclusively one dwelling including stairs, landings, etc., but excluding external balconies and edges of roofs (see C3)	0.35	0.35	0.6	0.5	0.25	→ N03R	
	Other residential (see C3)	0.75	0.75	0.6	1.0	0.5	→ N07R	
B, E Offices and work areas not included elsewhere including storage areas	Light access stairs and gangways not more than 600mm wide	0.22	0.22	0.6	N/A	N/A	→ N02	
	Fixed platforms, walkways, stairways and ladders for access	0.35	0.35	0.6	N/A	N/A	→ N03C	
	Areas not susceptible to overcrowding in office and institutional buildings also industrial and storage buildings	0.75	0.75	0.6	1.0	0.5	→ N07C	
AREAS WHERE PEOPLE MAY CONGREGATE								
C1/C2 Areas with tables or fixed seating	Areas with fixed seating adjacent to a balustrade, restaurants, bars. etc.	1.5	0.75	0.6	1.5	1.5	→ N15	
C3 Areas without obstacles for moving people not susceptible to over- crowding	Stairs, landings, external balconies, edges of roofs, etc.	0.75	0.75	0.6	1.0	0.5	→ N07C	
C5 Areas susceptible to over-crowding	Theatres, cinemas, grandstands, discheques, bars, clubs, auditoria, shopping malls (see also D), assembly areas, studios, etc	3.0*	0.75	0.6	1.5	1.5	N30	
							N20 ⁽¹⁾	
D Retail areas	All retails areas including public areas of banks/building societies (see C5 for areas where over- crowding may occur)	1.5	0.75	0.6	1.5	1.5	→ N15	

⁽¹⁾ N20 is only applicable where 2.0kN/m horizontal top edge load is acceptable by the Building Consent Authorities.

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